

Project Problem Statement - Final Version

PE6201 - Emerging AI Technologies | Nanyang Technological University

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Final revision of the original Week 3 proposal dated 23 August 2026. Updated to reflect the implemented Agent and completed evaluation.

1. Working title

A short, specific name for your project.

An Evidence-Grounded Singapore Tourism Agent with RAG and Controlled Tool Use

2. The problem, and why it matters

Who has this problem, and why is it worth solving?

Tourists must combine information from multiple official websites to check attraction hours, visitor rules, transport and accessibility. General-purpose chatbots can give plausible but unsupported answers, while static information can become outdated. This project addresses that problem by combining official-source retrieval with an Agent that can select tools, provide sources and evidence dates, and acknowledge insufficient evidence.

3. Who it is for, and the domain

Your user / audience and the industry.

The intended users are international tourists planning or taking a trip to Singapore, including families and travellers with accessibility needs. The domain is tourism and travel information. The prototype supports factual questions and evidence-based itinerary planning in English or Chinese; it does not book tickets, make payments or provide guaranteed live navigation.

4. Why AI - and which kind?

Is AI the right tool here, and which type?

The original proposal used a foundation model with RAG and excluded Agent tool calling. The final implementation extends that approach to a controlled Agent because different questions require different evidence operations. It combines semantic retrieval with four tools: `search_knowledge_base`, `get_route_information`, `fetch_live_official_page` and `search_live_official_web`. The model interprets natural-language questions, selects tools and synthesises evidence. Simple lookups remain suitable for fixed facts, but they cannot cover the same range of questions and multi-source planning.

5. Proposed approach - build vs buy or both

Which layers do you own or rent, and why?

I built the Python/Colab data pipeline, text extraction, chunking, multilingual embeddings, FAISS retrieval, official-domain controls, tool functions, audit logging and evaluation workflow. I used the OpenAI Agents SDK for orchestration and rented model inference through OpenRouter, comparing GPT-4o-mini with GPT-6.1. Retrieval used top-k 6 and a minimum similarity of 0.25. Adding live official web search as the fourth tool expanded current-source checking, but did not guarantee access to current prices or closure information. This build-and-buy approach allowed me to control evidence and evaluation without training a foundation model.

6. Data

What data is needed, where is it from, and what constraints apply?

The recorded official tourism snapshot is dated 4 October 2026. Sources include Gardens by the Bay, Mandai, Resorts World Sentosa, Sentosa, NParks, Visit Singapore and the National Heritage Board. Of 67 attempted documents, 35 were usable, 30 failed or provided insufficient text, and two required manual review. The usable corpus contained 66,605 text characters and 57 chunks. URLs, source metadata, dates, extraction status and checksums support provenance. No private user dataset was collected. Public availability does not establish unrestricted reuse rights; source attribution must be retained. Queries are processed through an external API.

Evaluation, Responsible Use and First Version

7. Success metric and evaluation

One metric tied to value, and an evaluation that can be run.

The original target was at least 85% factual consistency with source documents. The final evaluation used 30 manually designed questions, deterministic validation, LLM judging and ten human-reviewed items per model. The practical value criterion is a supported, correct answer that follows the question requirements. GPT-6.1 achieved 90% corrected tool-selection accuracy and 80% human overall pass, compared with 46.7% and 30% for GPT-4o-mini. GPT-6.1 human faithfulness was 80%, so the original 85% target was not demonstrated by this sample. Its validator pass rate was only 13.3%, showing that tool use, answer quality and formatting checks measure different things. The human-review subset is small, the judge changed between models, and GPT-4o-mini reviewed answers do not exactly match its final archived outputs; the comparison is indicative rather than fully controlled.

8. Risks, limitations and responsible use

What could go wrong, including silent failure, and who might it affect?

The main unresolved risk is stale or incomplete evidence. Dynamic booking pages can resist extraction, and an official URL or a successful tool call does not prove that its content answers the question. Even after adding the fourth tool, the Agent could not reliably obtain some real-time facts or align replies with the actual current date. It may cite an old closure notice or fail to confirm today's ticket price, potentially disrupting travel plans. Retrieved content is treated as untrusted data; official-domain restrictions, evidence dates, audit logs and abstention reduce risk but do not eliminate it. The prototype should provide qualified reference information, with current facts labelled verified only when applicable dated evidence is obtained.

9. Smallest first version (optional)

The tiniest end-to-end slice to build first.

The smallest version is one attraction and one factual query: load an official Gardens by the Bay snapshot, create embeddings and a FAISS index, retrieve the relevant opening-hours passage, and generate an answer with the official URL and snapshot date. Check whether the answer is supported and whether it distinguishes hours across attractions. Expand this slice to the four-tool Agent, the broader corpus and the 30-question evaluation. A future version should add dependable Singapore-time handling, improved extraction of dynamic pages, and a fixed independent evaluator before claiming reliable real-time assistance.